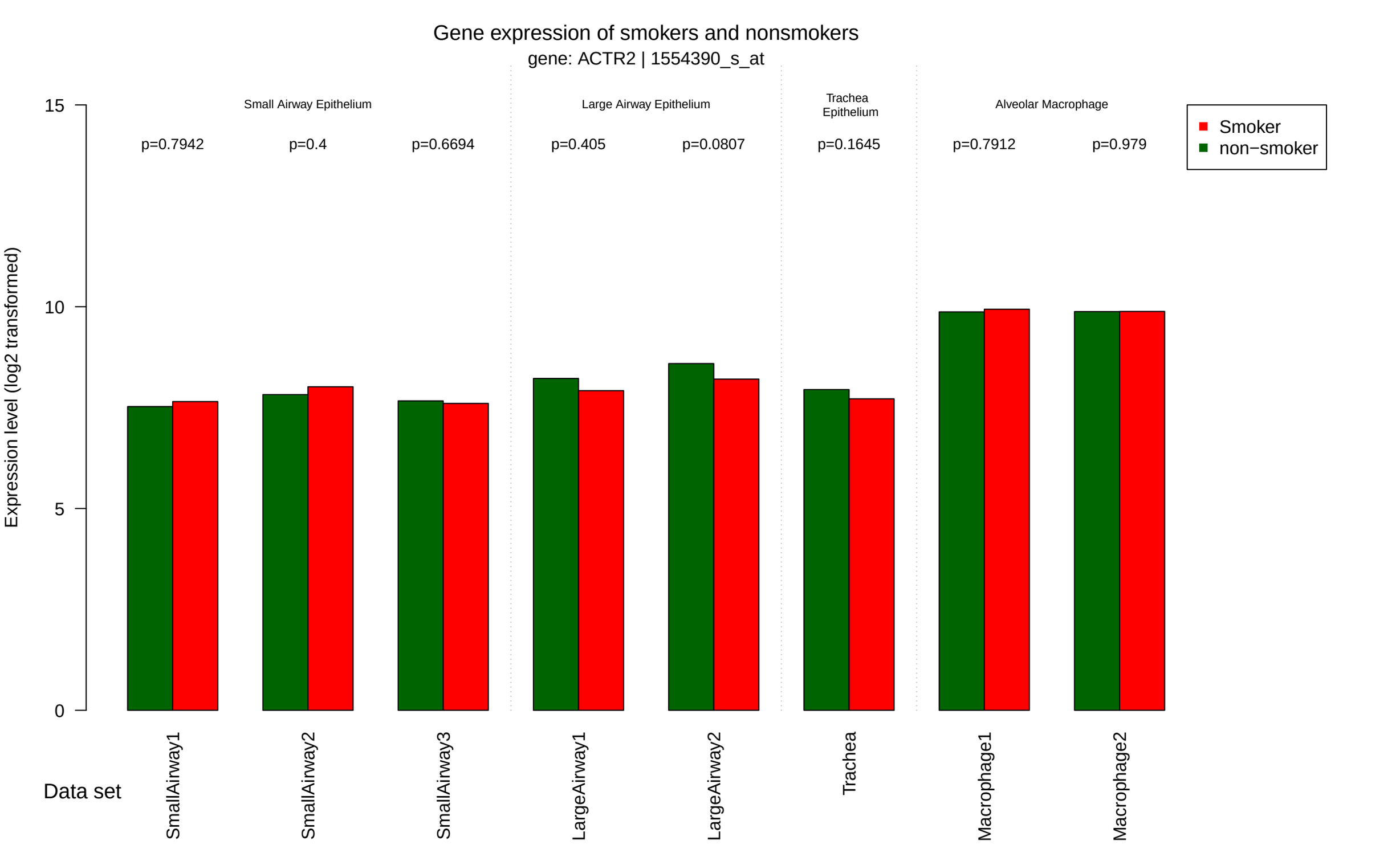
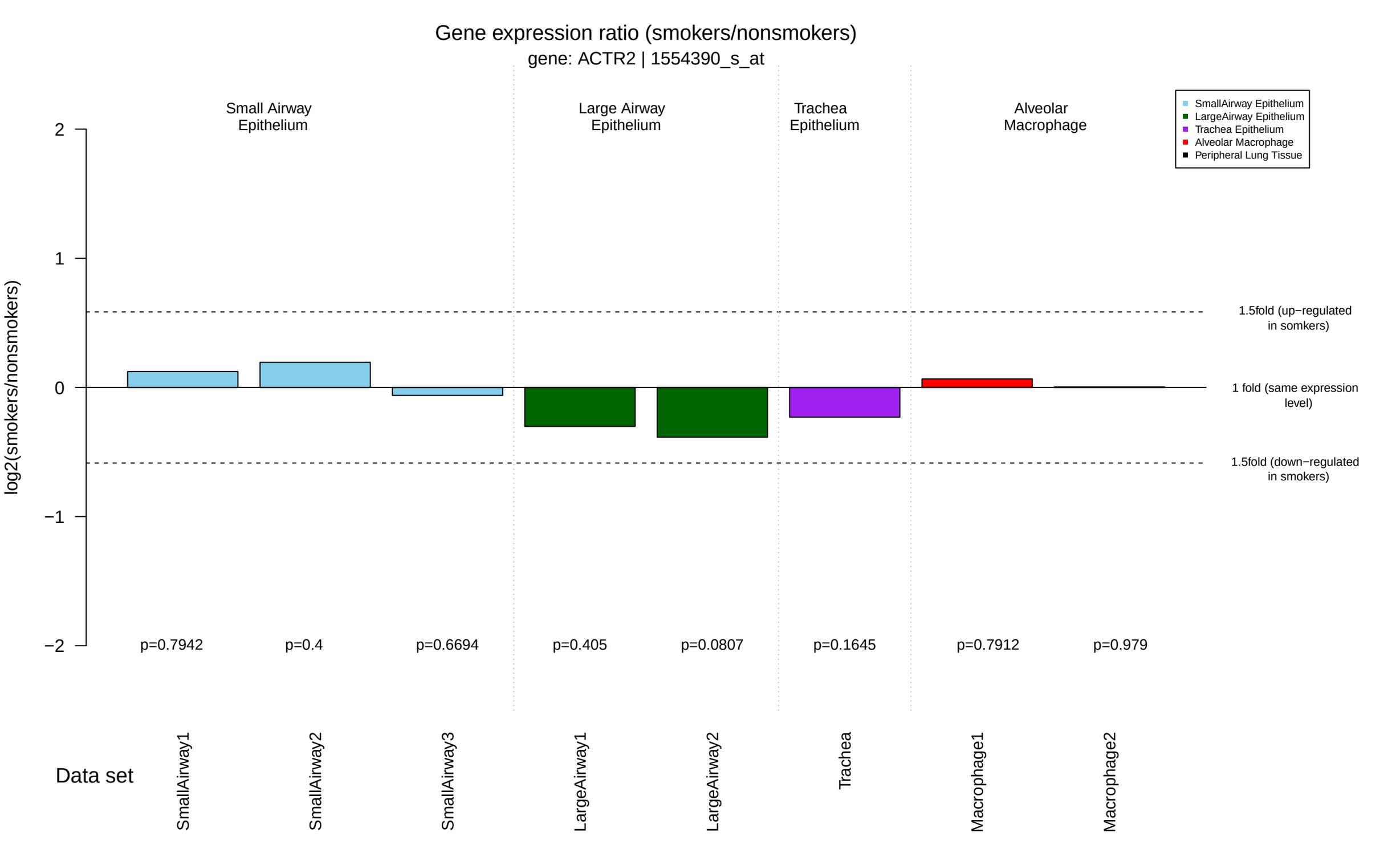
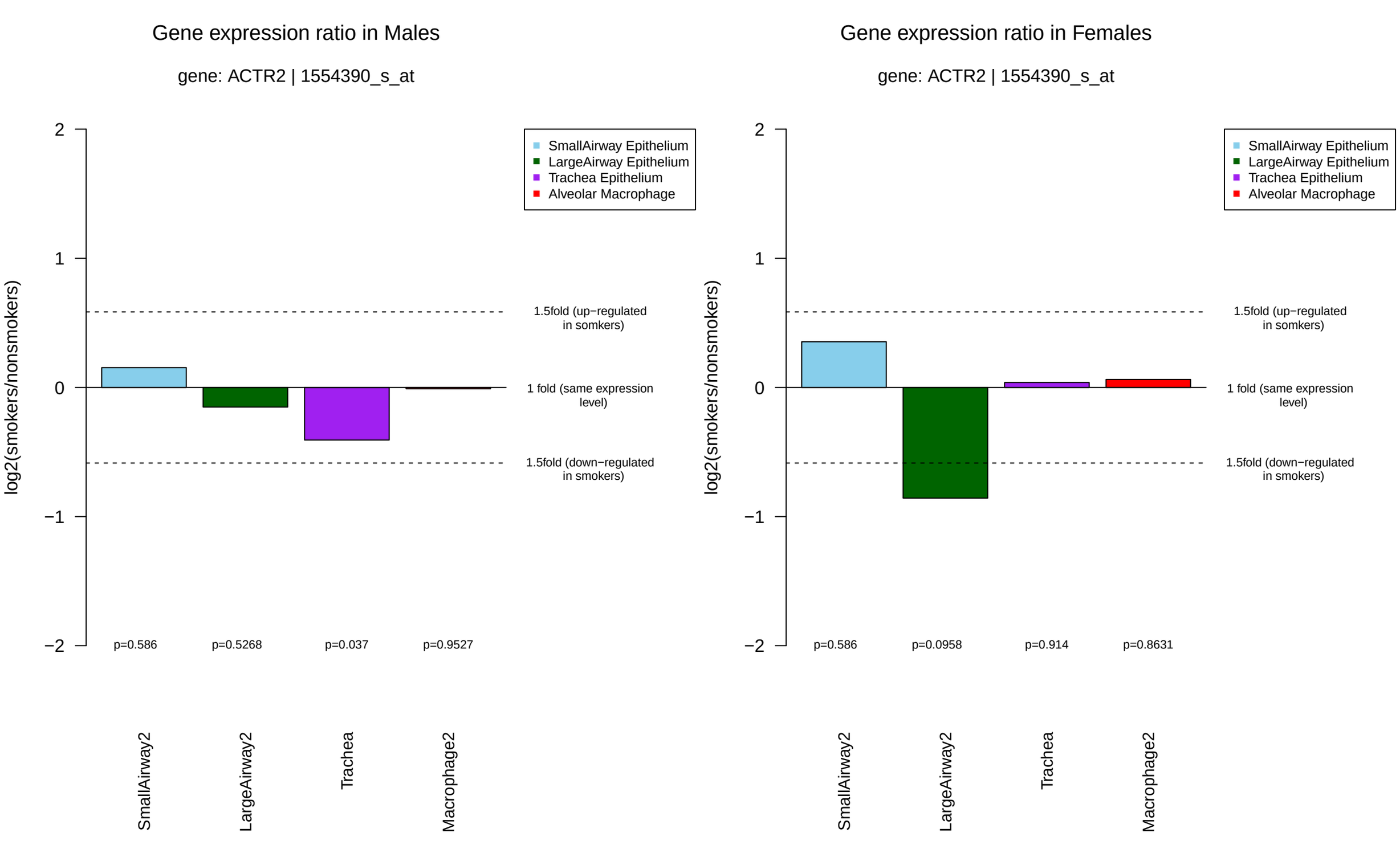
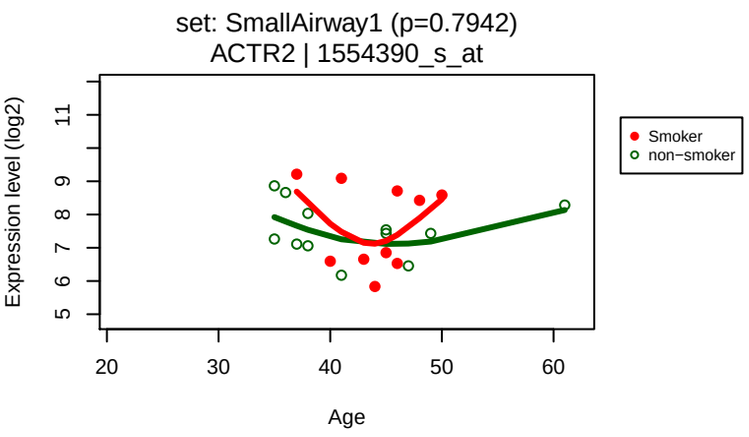
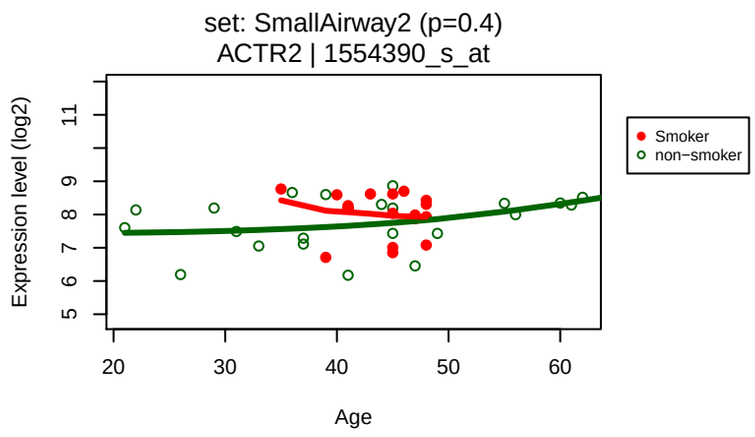
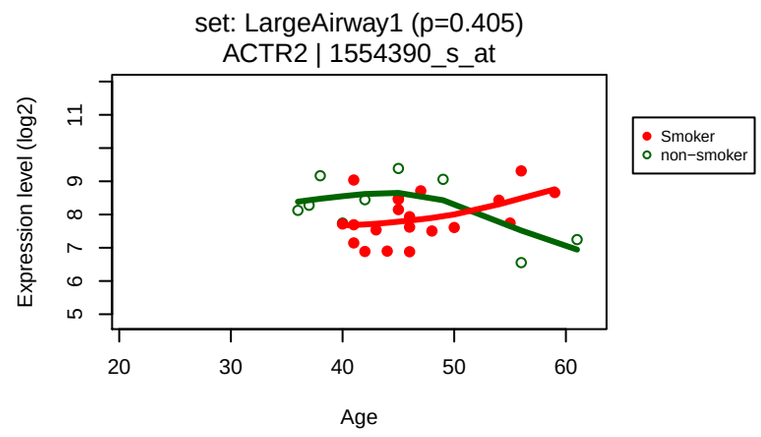

Gene expression of smokers and nonsmokers
gene: ACTR2 | 1554390_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7942
p=0.4
p=0.6694
p=0.405
p=0.0807
p=0.1645
p=0.7912
p=0.979
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

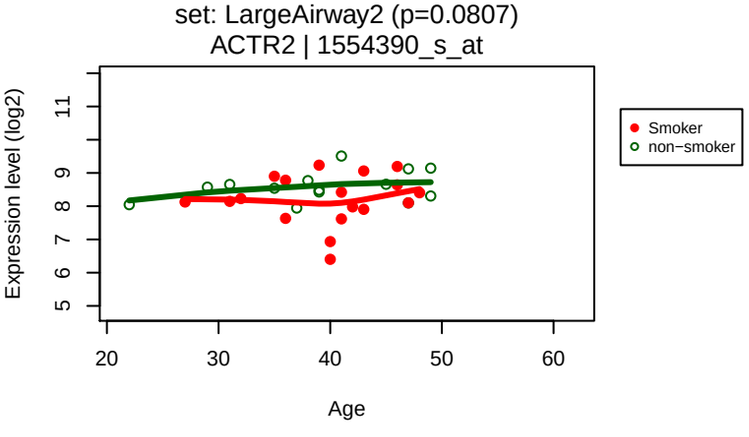

Gene expression ratio (smokers/nonsmokers)
gene: ACTR2 | 1554390_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7942
p=0.4
p=0.6694
p=0.405
p=0.0807
p=0.1645
p=0.7912
p=0.979
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

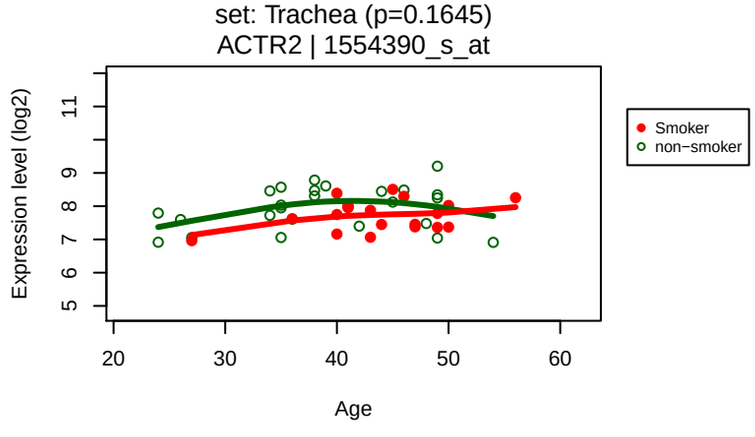

Gene expression ratio in Males
gene: ACTR2 | 1554390_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.586
p=0.5268
p=0.037
p=0.9527
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTR2 | 1554390_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.586
p=0.0958
p=0.914
p=0.8631
SmallAirway2
LargeAirway2
Trachea
Macrophage2

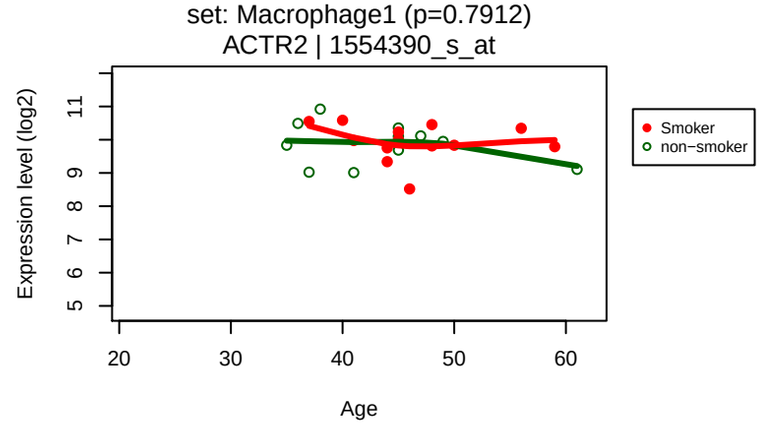

set: SmallAirway1 (p=0.7942)
ACTR2 | 1554390_s_at
Expression level (log2)
Age
Smoker
non−smoker

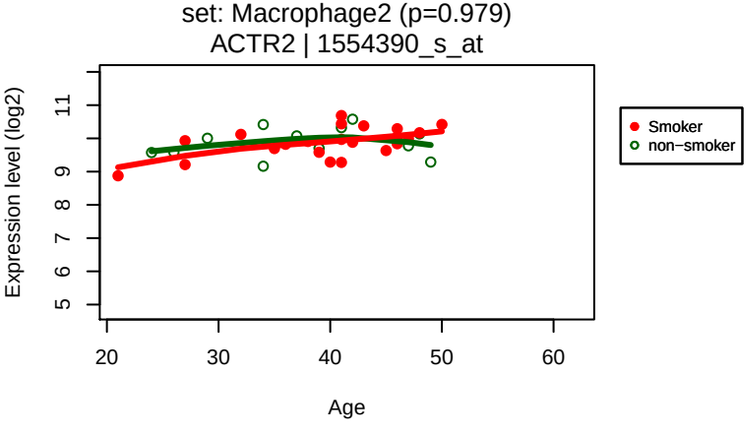

set: SmallAirway2 (p=0.4)
ACTR2 | 1554390_s_at
Expression level (log2)
Age
Smoker
non−smoker

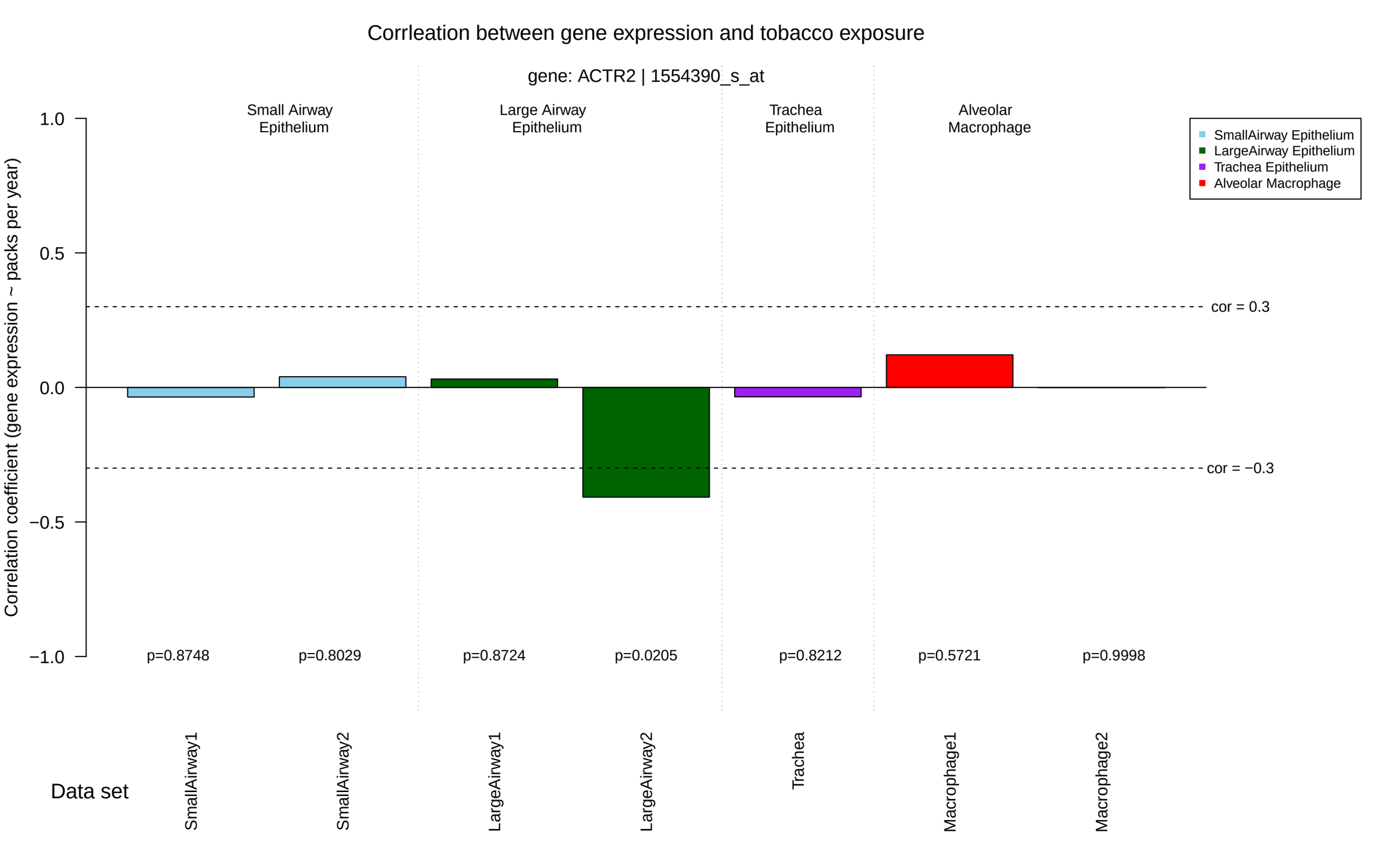

set: LargeAirway1 (p=0.405)
ACTR2 | 1554390_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0807)
ACTR2 | 1554390_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1645)
ACTR2 | 1554390_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7912)
ACTR2 | 1554390_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.979)
ACTR2 | 1554390_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTR2 | 1554390_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8748
p=0.8029
p=0.8724
p=0.0205
p=0.8212
p=0.5721
p=0.9998
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2